Modality and dynamical systems theory

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MUCH of what we notice around us are not not things that are, but rather changes in what there is. Perhaps you wake up one morning while the news declares that the central bank in your country has found that the inflation has increased too much, and that the bank must increase the interest rate. That could make people less inclined to buy your house, and that could, in turn, by the law of demand and supply, lead to a decreased value of your property. This in turn could give you problem to pay off your mortgage when moving.

Suppose you pull the blinds and go to bed again.

All of this is about change. So is climate change (you realize that directly just by seeing the concept), species extinctions and decrease in biodiversity. The fall of republics, and the rise of empires. Revolutions. Innovations and invasions. You name it. There is a tradition of Western philosophy which studies processes. It goes from Heraclitus of Ephesus (born ca. 560 B.C.E.), to modern *process philosophy* via (for example) Georg Wilhelm Friedrich Hegel, Karl Marx, Alfred North Whitehead, Martin Heidegger and Wilfred Sellars (Seibt, 2025). To tackle processes, we need to concentrate more on what is becoming than on what there is.

However, I leave most process philosophy here. In this note I concentrate on subset of change, namely those processes that are studied using methodologies from dynamical systems theory.[1] I aim at elucidating the modal free lunch we get from dynamical systems (Maudlin, 2020). I accept that lunch, and pay with a few lines on causality and laws of nature. Then I discuss modality from the points of view of Quine and Kripke: What changes has to be contingent. The change itself has to be possible. Here I discuss on what there need to be for us study the becoming, that is the dynamical systems.

Then I turn to the modal sciences and spaces of possibility (Williamson, 2016; Williamson, 2018). Here I concentrate at the analogy between orbits or trajectories and possible worlds semantics. I also analyse the state variables of dynamical systems in terms of ontological commitments and arguments for and agains essentialism. Even if science may need generous allocations of natural kinds and biological species, It is not obvious that it supports essentialism. Where applicable, I give examples from population biology and evolution with systems with interesting properties. After that it is time to draw some conclusions.

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We expect or fear changes, and talk about the corresponding modalities with adverbs and auxiliary verbs like

can, can not, could ...

may, might, must ...

able to, has to, needs to ...

All of them are related to possibilities, contingencies and necessities. There are further distinctions (Kment, 2021), such as

[1] A branch of mathematics, working with difference and differential equations to model processes. I learned it from Luenberger (1979).

Metaphysical or *objective modality:* things that are necessary or possible in the real world.

Nomic modality: a special case of metaphysical modality which is related to necessities and contingencies that are entailed from the laws of nature.

Epistemic or *subjective modality:* related to whether we know something for sure or just believe it. Subjective modality seems to be a special case of epistemic modality related to the strengths of beliefs.

At its meeting 2006 the International Astronomical Union redefined Pluto from a planet to a dwarf planet. *And then they were only eight*. As a matter of fact, Pluto is a more complex celestial body than most of us realise. It is a dwarf planet with five moons, of which the largest is almost as large as the planet itself. In a poor telescope it appears much larger than it is (Wikipedia, 2026). Since that decision was controversial, some of us would perhaps still say

(P1) Possibly ⌜the number planets = 9⌝

After all, even before 2006 we would say that the number of planets is *contingent*. It could change. Astronomers could direct their telescopes towards far end of the solar system and possibly discover yet another. That had happened before. Note how close it is between metaphysical and epistemic modalithy.

I am old enough to have learned in school that

(P2) the number planets = 9

and also that

(P3) 9 is *necessarily* greater than 7

Substituting statement P2 into P3 yields

(P4) the number of planets is *necessarily* greater than 7

which is false, according to what we agreed upon in P1. What is possible cannot be necessary. According to Quine (1947) quantified referential modal logic can yield nonsense. He points out that

$$\forall\, x\, \forall\, y \Big[(x = y) \rightarrow \Box (x = y) \Big] \tag{1}$$

can only be true if $x = y$ is an essential property of x and y. Eq. (1) is a truth when $x = y$ is true *a priori*, which in addition is the same as accepting *analyticity*. Eq. (1) can be as false as proposition (P4). We will not see the reference x and y because of *referential opacity*. Quine claims that this leads to queer ontologies:

> What I shall show is that it has queer ontological consequences. It leads us to hold that there are no concrete objects (men, planets, etc.), but rather that there are only, corresponding to each supposed concrete object, a multitude of distinguishable entities (perhaps "individual concepts," in Church's phrase). It leads us to hold, e. g., that there is no such ball of matter as the so-called planet Venus, but rather at least three distinct entities: Venus, Evening Star, and Morning Star (Quine, 1947).

Here is where the referential opacity comes in, as something between the reference and the terms. We can not see that ball of matter, hence there is Venus and the Evening Star and the Morning Star, each on its own. Or as Kripke puts it:

> Quine has also urged that even if these objections have been answered, modal logic would clearly commit itself to the metaphysical jungle of Aristotelian essentialism (Kripke, 2017).

Quine himself claims that it requires us to adopt Aristotelian essentialism:

> Things had essences, for Aristotle, but only linguistic forms have meanings. Meaning is what essence becomes when it is divorced from the object of reference and wedded to the word (Quine, 1951).

Which is exactly what Quine does not want. I will now discuss Kripke's criticism of Quine. Then further on I will offer you a guided tour into that metaphysical jungle we might be forced to cope with.

Kripke first introduces a distinction between *pure* and *applied* semantics. The two variants of semantics is very much like pure and applied mathematics or biology or whatever.

Modal reasoning appears everywhere in science says Timothy Williamson. He claims that dynamical systems theory, which is abundantly used in natural sciences, is by its very nature modal, in spite of the fact that these sciences hardly ever use modal logic *per se* (Williamson, 2018). He gives a much more detailed treatment in his paper "Modal science" (Williamson, 2016). Tim Maudlin shares Williamson's views on modality in the sciences (Maudlin, 2020), but does so from the point of view of the philosophy of physics.

Objective and subjective modalities are Williamson's choice of terminology. He argues that this mirrors the vocabulary used for probability, where objective probability refers to truly stochastic processes and subjective probability refers to the strength of a conviction. I will follow him on this in using objective and subjective as determinants for both probabilities and modalities.

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